

EXAMPLE 1. Solve the equation $x^2 - 7x + 12 = 0$, and find the value of x .

- a. $-3, -4$ b. $-3, 4$ c. $3, -4$ d. $3, 4$

Sol. d. Given, $x^2 - 7x + 12 = 0$
 $x^2 - 3x - 4x + 12 = 0$ [splitting the middle term]
 $\Rightarrow x(x - 3) - 4(x - 3) = 0$
 $\Rightarrow (x - 3)(x - 4) = 0$
 $\Rightarrow (x - 3) = 0$ or $(x - 4) = 0 \Rightarrow x = 3$ or $x = 4$
 So, $x = 3, 4$ are roots of the given equation.

EXAMPLE 2. Solve the equation $3a^2x^2 + 8abx + 4b^2 = 0$, $a \neq 0$, and find the value of x .

- a. $\frac{2b}{3a}, \frac{2b}{a}$ b. $\frac{-2b}{3a}, \frac{2b}{a}$ c. $\frac{-2b}{3a}, \frac{-2b}{a}$ d. $\frac{2b}{3a}, \frac{-2b}{a}$

Sol. c. The equation is $3a^2x^2 + 8abx + 4b^2 = 0$
 $\Rightarrow 3a^2x^2 + 2abx + 6abx + 4b^2 = 0$
 $\Rightarrow ax(3ax + 2b) + 2b(3ax + 2b) = 0$
 $\therefore (3ax + 2b)(ax + 2b) = 0$
 $\Rightarrow 3ax + 2b = 0$ or $ax + 2b = 0$
 $\therefore x = \frac{-2b}{3a}$ or $\frac{-2b}{a}$

2. By Using the Quadratic Formula

Let the quadratic equation be $ax^2 + bx + c = 0$, where $a \neq 0$. Then, solution of this equation can be found using the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- If α and β be considered as roots of the quadratic equation, then

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

- The quantity $b^2 - 4ac$ is called the discriminant of the quadratic equation $ax^2 + bx + c = 0$ and is denoted by D . So, $D = b^2 - 4ac$.

EXAMPLE 3. Solve the equation $x^2 - 9x + 18 = 0$ and find the value of x .

- a. $3, 6$ b. $-3, -6$ c. $-3, 6$ d. $3, -6$

Sol. a. We have, $x^2 - 9x + 18 = 0$
 Here, $a = 1$, $b = -9$ and $c = 18$
 $\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(18)}}{2(1)}$
 $= \frac{9 \pm \sqrt{81 - 72}}{2} = \frac{9 \pm \sqrt{9}}{2} = \frac{9 \pm 3}{2}$
 $\therefore x = \frac{9 + 3}{2}$ or $x = \frac{9 - 3}{2} \Rightarrow x = 6$ or 3
 Hence, the values of x are 6 and 3.

Nature of Roots of Quadratic Equation

Let $D = b^2 - 4ac$ be the discriminant of the quadratic equation, $ax^2 + bx + c = 0$, where $a \neq 0$. Then, the following cases arise

- If $D > 0$, then the two roots are real and distinct.
- If $D = 0$, then the two roots are real and equal and are given by $\alpha = \beta = \frac{-b}{2a}$.
- If $D < 0$, then there are no real roots, i.e. given equation has imaginary roots.

IMPORTANT POINTS

- If $a, b, c \in \mathbb{Q}$ and D is a perfect square, then equation has rational roots.
- If one root of quadratic equation is $p + \sqrt{q}$, then other root will be $p - \sqrt{q}$.

EXAMPLE 4. If the equation $x^2 + 2(1+k)x + k^2 = 0$ has equal roots, then what is the value of k ?

- a. $\frac{1}{2}$ b. $-\frac{1}{2}$
 c. 1 d. -1

Sol. b. Given equation, $x^2 + 2(1+k)x + k^2 = 0$.

If it has equal roots, then $D = 0$

$$\Rightarrow \{2(1+k)\}^2 - 4k^2 = 0$$

$$\Rightarrow 4(1+k^2 + 2k) - 4k^2 = 0$$

$$\Rightarrow 4 + 4k^2 + 8k - 4k^2 = 0$$

$$\Rightarrow 4 + 8k = 0 \Rightarrow k = -\frac{4}{8}$$

$$\therefore k = -\frac{1}{2}$$

Hence, the value of k is $-\frac{1}{2}$.

Roots Under Particular Conditions

Consider the quadratic equation $ax^2 + bx + c = 0$.

- Both roots are positive, if a and b are opposite in sign and a and c are same in sign, i.e. $\frac{-b}{a} > 0$ and $\frac{c}{a} > 0$.
- Both roots are negative, if a, b and c are of same sign, i.e. $\frac{b}{a} > 0$ and $\frac{c}{a} > 0$.
- Roots are of opposite signs, if a and c are of opposite sign, i.e. $\frac{-a}{c} > 0$.
- Roots are equal but opposite in signs, if $b = 0$.
- Roots are reciprocal to each other, if $a = c$.

EXAMPLE 5. What is the least integral value of k for which the equation $x^2 - 2(k-1)x + (2k+1) = 0$ has both the roots positive?

- a. 1 b. $-\frac{1}{2}$ c. 4 d. 0

Sol. a. Both the roots of the equation $ax^2 + bx + c = 0$ are positive, if $-\frac{b}{a} > 0$ and $\frac{c}{a} > 0$

Given equation is $x^2 - 2(k-1)x + (2k+1) = 0$.

Now, $-\frac{b}{a} = \frac{2(k-1)}{1} \Rightarrow 2(k-1) > 0$, if $k > 1$

and $\frac{c}{a} = \frac{2k+1}{1} \Rightarrow (2k+1) > 0$, if $k > -\frac{1}{2}$

$\therefore k > 1$

Hence, the least value k as per given in options is 1.

Relation between Roots and Coefficients

1. Quadratic Equation

Let α, β be the roots of the equation $ax^2 + bx + c = 0$ ($a \neq 0$), then

Sum of the roots $= \alpha + \beta = -\frac{b}{a} = \frac{-\text{Coefficient of } x}{\text{Coefficient of } x^2}$

Product of the roots $= \alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$

2. Cubic Equation

If α, β, γ are the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$, then

Sum of roots $= \alpha + \beta + \gamma = -b/a$

Product of roots two at a time $= \alpha\beta + \beta\gamma + \gamma\alpha = c/a$

Product of three roots $= \alpha\beta\gamma = -\frac{d}{a}$

EXAMPLE 6. What are the roots of the quadratic equation $a^2b^2x^2 - (a^2 + b^2)x + 1 = 0$?

- a. $\frac{1}{a^2}, \frac{1}{b^2}$ b. $-\frac{1}{a^2}, -\frac{1}{b^2}$ c. $\frac{1}{a^2}, -\frac{1}{b^2}$ d. $-\frac{1}{a^2}, \frac{1}{b^2}$

Sol. a. Let roots of the equation $a^2b^2x^2 - (a^2 + b^2)x + 1 = 0$ be α and β .

$\therefore$ Sum of roots $= \alpha + \beta = \frac{a^2 + b^2}{a^2b^2}$... (i)

and product of roots $= \alpha\beta = \frac{1}{a^2b^2}$

Now, $\alpha - \beta = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\left(\frac{a^2 + b^2}{a^2b^2}\right)^2 - \frac{4}{a^2b^2}}$

$$\Rightarrow \alpha - \beta = \sqrt{\frac{(a^2 - b^2)^2}{(a^2b^2)^2}} = \frac{a^2 - b^2}{a^2b^2} \quad \dots(ii)$$

On solving Eqs. (i) and (ii), we get

$$\alpha = \frac{1}{b^2} \text{ and } \beta = \frac{1}{a^2}$$

EXAMPLE 7. If one root of the equation $\frac{x^2}{a} + \frac{x}{b} + \frac{1}{c} = 0$ is reciprocal to the other, then which one of the following is correct?

- a. $a = b$ b. $b = c$ c. $ac = 1$ d. $a = c$

Sol. d. Given, $\frac{x^2}{a} + \frac{x}{b} + \frac{1}{c} = 0$... (i)

Now, let one root be α , then the other root is $1/\alpha$.

We know that,

$$\text{Product of roots} = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = \frac{1/c}{1/a}$$

$$\Rightarrow \alpha \cdot \frac{1}{\alpha} = \frac{a}{c} \Rightarrow c = a$$

which is the required relation.

Symmetric Functions of the Roots

Let α and β be the roots of a quadratic equation. An expression in α and β which remains same when α and β are interchanged is known as a symmetric function in α and β . To evaluate the value of symmetric function of the roots, express the given function in terms of $(\alpha + \beta)$ and $\alpha\beta$.

You can use the following results.

- $(\alpha^2 + \beta^2) = [(\alpha + \beta)^2 - 2\alpha\beta]$
- $(\alpha^3 + \beta^3) = [(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)]$
- $(\alpha - \beta)^2 = [(\alpha + \beta)^2 - 4\alpha\beta]$
- $(\alpha^3 - \beta^3) = [(\alpha - \beta)^3 + 3\alpha\beta(\alpha - \beta)]$

EXAMPLE 8. If α and β are the roots of the equation $x^2 - 6x + 6 = 0$, what is $\alpha^3 + \beta^3 + \alpha^2 + \beta^2 + \alpha + \beta$ equal to?

- a. 150 b. 138 c. 128 d. 124

Sol. b. Here, $\alpha + \beta = \frac{-\text{Coefficient of } x}{\text{Coefficient of } x^2} = 6$

and $\alpha\beta = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = 6$

$$\therefore (\alpha + \beta)^2 = 6^2 \Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = 36$$

$$\Rightarrow \alpha^2 + \beta^2 = 36 - 2(6) = 24$$

$$\begin{aligned} \therefore (\alpha^3 + \beta^3) + (\alpha^2 + \beta^2) + (\alpha + \beta) &= (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta) + (\alpha^2 + \beta^2) + (\alpha + \beta) \\ &= 6(24 - 6) + (24) + (6) \\ &= 6(18) + 30 = 108 + 30 = 138 \end{aligned}$$

Formation of a Quadratic Equation

If α, β are the roots of a quadratic equation, then the equation is $(x-\alpha)(x-\beta)=0 \Rightarrow x^2-(\alpha+\beta)x+\alpha\beta=0$
i.e. $x^2 - (\text{sum of the roots})x + \text{product of the roots} = 0$

EXAMPLE 9. The quadratic equation whose roots are 3 and -1 , is

- a. $x^2 - 4x + 3 = 0$ b. $x^2 - 2x - 3 = 0$
c. $x^2 + 2x - 3 = 0$ d. $x^2 + 4x + 3 = 0$

Sol. b. Given that, the roots of the quadratic equation are 3 and -1 .

Let $\alpha = 3$ and $\beta = -1$
 $\therefore$ Sum of roots $= \alpha + \beta = 3 - 1 = 2$
and product of roots $= \alpha\beta = (3)(-1) = -3$
So, required quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$
$$\Rightarrow x^2 - (2)x + (-3) = 0$$
$$\Rightarrow x^2 - 2x - 3 = 0$$

Equations Reducible to Quadratic Equations

The equations which at the outset are not quadratic equations, but can be reduced to quadratic equations by suitable substitutions or simplifications are called equations reducible to quadratic equations.

(Rule 1) $ax^{2n} + bx^n + c = 0$, where $n \geq 2$

Use substitution $x^n = y$

$\Rightarrow x^{2n} = y^2$ and equation reduces to $ay^2 + by + c = 0$.

EXAMPLE 10. Solve the quadratic equation $x^4 - 26x^2 + 25 = 0$, and find the value of x .

- a. $\pm 1, \pm 25$ b. $\pm 1, \pm 5$
c. $1, 5$ d. $1, 25$

Sol. b. Put $x^2 = z \Rightarrow x^4 = z^2$

$$x^4 - 26x^2 + 25 = 0 \text{ can be written as}$$
$$z^2 - 26z + 25 = 0 \Rightarrow z^2 - 25z - z + 25 = 0$$
$$\Rightarrow z(z - 25) - 1(z - 25) = 0$$
$$\Rightarrow (z - 1)(z - 25) = 0 \Rightarrow z = 1 \text{ or } z = 25$$

when $z = 1 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$
when $z = 25 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$
Hence, the solutions are ± 1 and ± 5 .

(Rule 2) $Px + \frac{Q}{x} = R$

Reduce the given equation to a quadratic equation by multiplying both sides by ' x '. So, $Px^2 + Q = Rx$

or $Px^2 - Rx + Q = 0$

EXAMPLE 11. Solve $2x - \frac{3}{x} = 5$. Then, the value of x is

- a. $1/3, -3$ b. $1/2, -3$ c. $-1/2, 3$ d. $1/2, 4$

Sol. c. Here, $2x - \frac{3}{x} = 5$... (i)

On multiplying Eq. (i) by $x \Rightarrow 2x^2 - 3 = 5x$
 $\Rightarrow 2x^2 - 5x - 3 = 0 \Rightarrow 2x^2 - 6x + x - 3 = 0$
 $\Rightarrow (2x + 1)(x - 3) = 0 \Rightarrow x = -1/2 \text{ or } x = 3$
Hence, the solution is $x = \frac{-1}{2}, 3$.

(Rule 3) $\sqrt{x+b} + x = c$ or $\sqrt{a-x^2} = bx + c$

- Square both sides to get a quadratic equation without the radical.
- Solve the equation by factorisation or by quadratic formula.

EXAMPLE 12. Solve $\sqrt{2x+9} + x = 13$, then the value of x is

- a. $4, 10$ b. $8, 20$ c. $3, 15$ d. $-4, -10$

Sol. b. Given, $\sqrt{2x+9} + x = 13 \Rightarrow \sqrt{2x+9} = 13 - x$

On squaring both sides, we get $(\sqrt{2x+9})^2 = (13-x)^2$
 $\Rightarrow 2x+9 = 169 + x^2 - 26x$
 $\Rightarrow x^2 - 28x + 160 = 0 \Rightarrow x^2 - 20x - 8x + 160 = 0$
 $\Rightarrow (x-8)(x-20) = 0 \Rightarrow x = 8 \text{ or } x = 20$
Hence, the solution is $x = 8, 20$.

(Rule 4) $\sqrt{ax+b} \pm \sqrt{cx+d} = e$

or $\sqrt{ax+b} \pm \sqrt{cx+d} \pm \sqrt{ex+f} = 0$

To solve such type of equations, follow the steps given below.

- Square both sides once, so that only one term containing radical is obtained.
- Keep the term containing radical on one side and all other terms on the other side.
- Square again to get, the quadratic equation and solve it.

EXAMPLE 13. Solve the following equation

$\sqrt{4-x} + \sqrt{x+9} = 5$. Then, the value of x is

- a. $1, 5$ b. $-1, 3$ c. $0, -5$ d. $1, 4$

Sol. c. Given, $\sqrt{4-x} + \sqrt{x+9} = 5$

On squaring both sides, we get $(\sqrt{4-x} + \sqrt{x+9})^2 = 25$
 $\Rightarrow (4-x) + (x+9) + 2(\sqrt{4-x}\sqrt{x+9}) = 25$
 $\Rightarrow 2(\sqrt{4-x})(\sqrt{x+9}) = 12 \Rightarrow (\sqrt{4-x})(\sqrt{x+9}) = 6$
Again, squaring both sides, we get
 $\Rightarrow (4-x)(x+9) = 36 \Rightarrow -x^2 + 36 - 5x = 36$
 $\Rightarrow x^2 + 5x = 0 \Rightarrow x(x+5) = 0 \Rightarrow x = 0 \text{ or } x = -5$
 $\therefore x = 0, -5$ is the required solution.

EXAMPLE 14. Solve $\sqrt{11y-6} + \sqrt{y-1} - \sqrt{4y+5} = 0$.
Then, the value of y is

- a. {5, 6/5} b. {1, 2}
c. {3, 5/2} d. None of these

Sol. a. Given, $\sqrt{11y-6} + \sqrt{y-1} - \sqrt{4y+5} = 0$

$$\sqrt{11y-6} + \sqrt{y-1} = \sqrt{4y+5}$$

On squaring both sides, we get

$$[\sqrt{11y-6} + \sqrt{y-1}]^2 = [\sqrt{4y+5}]^2$$

$$\Rightarrow (11y-6) + (y-1) + 2\sqrt{(11y-6)(y-1)} = (4y+5)$$

$$\Rightarrow 12y-7 + 2\sqrt{11y^2-17y+6} = 4y+5$$

$$\Rightarrow 2\sqrt{11y^2-17y+6} = 12-8y$$

$$\Rightarrow \sqrt{11y^2-17y+6} = 6-4y$$

Again, squaring both sides, we get

$$11y^2-17y+6 = (6-4y)^2$$

$$\Rightarrow 11y^2-17y+6 = 36+16y^2-48y$$

$$\Rightarrow 5y^2-31y+30=0$$

Here, $a=5$, $b=-31$ and $c=30$

$$\Rightarrow y = \frac{31 \pm \sqrt{(-31)^2 - 4(5)(30)}}{2 \times 5} \left[\because y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right]$$

$$\Rightarrow y = \frac{31 \pm \sqrt{361}}{10} \Rightarrow y = \frac{31 \pm 19}{10}$$

$$\Rightarrow y = \frac{31+19}{10} \text{ and } y = \frac{31-19}{10}$$

$$\therefore y = \left\{ 5, \frac{6}{5} \right\} \text{ is the required solution.}$$

(Rule 5) $(x+a)(x+b)(x+c)(x+d)+k=0$, where

$a+b=c+d$ and k may or may not be zero.

To solve such type of equations, put $x^2+(a+b)x=T$.

EXAMPLE 15. Solve the equation

$(x+1)(x+2)(x+3)(x+4)-8=0$, and find the number of real roots of this equation.

- a. 1 b. 2 c. 4 d. No real roots

Sol. b. $(x+1)(x+2)(x+3)(x+4)-8=0$

Here, $1+4=2+3$,

$$\text{so, } [(x+1)(x+4)][(x+2)(x+3)]-8=0$$

$$[x^2+5x+4][x^2+5x+6]-8=0$$

Put $x^2+5x=t$, so the equation becomes

$$(t+4)(t+6)-8=0$$

$$\Rightarrow t^2+10t+24-8=0 \Rightarrow t^2+10t+16=0$$

$$\Rightarrow t^2+8t+2t+16=0$$

$$\Rightarrow t(t+8)+2(t+8)=0 \Rightarrow (t+8)(t+2)=0$$

$$\Rightarrow t = -8 \text{ or } -2$$

Now,

$$\begin{array}{l|l} x^2+5x=-8 & x^2+5x=-2 \\ \Rightarrow x^2+5x+8=0 & \Rightarrow x^2+5x+2=0 \\ \Rightarrow x = \frac{-5 \pm \sqrt{25-32}}{2} & \Rightarrow x = \frac{-5 \pm \sqrt{25-8}}{2} \\ \therefore x = \frac{-5 \pm \sqrt{-7}}{2} & \therefore x = \frac{-5 \pm \sqrt{17}}{2} \end{array}$$

Hence, the number of real roots of the given equation is 2.

(Rule 6) (i) $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x + \frac{1}{x}\right) + c = 0$

(ii) $a\left(x^2 + \frac{1}{x^2}\right) + b\left(x - \frac{1}{x}\right) + c = 0$

To solve an equation of the form (i) .

put $\left(x + \frac{1}{x}\right) = y$ and to solve an equation of

the form (ii), put $\left(x - \frac{1}{x}\right) = y$

EXAMPLE 16. Solve $4\left(x^2 + \frac{1}{x^2}\right) - 4\left(x + \frac{1}{x}\right) - 7 = 0$;

$x \neq 0$ and find the real roots of x .

- a. $\{-2, -1/2\}$ b. $\{0, 2\}$
c. $\{2, 1/2\}$ d. None of these

Sol. c. $4\left(x^2 + \frac{1}{x^2}\right) - 4\left(x + \frac{1}{x}\right) - 7 = 0$... (i)

Put $x + \frac{1}{x} = y$

On squaring both sides, we get

$$\left(x + \frac{1}{x}\right)^2 = y^2 \Rightarrow x^2 + \frac{1}{x^2} + 2 = y^2 \Rightarrow x^2 + \frac{1}{x^2} = y^2 - 2$$

$$\Rightarrow 4(y^2 - 2) - 4y - 7 = 0 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow 4y^2 - 8 - 4y - 7 = 0 \Rightarrow 4y^2 - 4y - 15 = 0$$

$$\Rightarrow 4y^2 - 10y + 6y - 15 = 0 \Rightarrow 2y(2y - 5) + 3(2y - 5) = 0$$

$$\Rightarrow (2y - 5)(2y + 3) = 0 \Rightarrow y = \frac{5}{2} \text{ or } -\frac{3}{2}$$

Now, For $y = \frac{5}{2}$ For $y = -\frac{3}{2}$

$$\Rightarrow x + \frac{1}{x} = \frac{5}{2} \quad \Rightarrow x + \frac{1}{x} = -\frac{3}{2}$$

$$\Rightarrow \frac{x^2 + 1}{x} = \frac{5}{2} \quad \Rightarrow \frac{x^2 + 1}{x} = -\frac{3}{2}$$

$$\Rightarrow 2x^2 + 2 = 5x \quad \Rightarrow 2x^2 + 2 = -3x$$

$$\Rightarrow 2x^2 - 5x + 2 = 0 \quad \Rightarrow 2x^2 + 3x + 2 = 0$$

$$\Rightarrow (x-2)(2x-1) = 0 \quad \text{Here, } D = 9 - 16 = -7 < 0$$

$$\therefore x = 2 \text{ or } x = \frac{1}{2} \quad \text{So, there are no real roots.}$$

$$[\because D = b^2 - 4ac]$$

Hence, the values of x are 2 and $\frac{1}{2}$.

Rule 7 $ax^4 + bx^3 + cx^2 + bx + a = 0$

- To solve such type of equations, we divide the equation by x^2 , to obtain $a(x^2 + 1/x^2) + b(x + 1/x) + c = 0$, and, then put $x + \frac{1}{x} = y$

EXAMPLE 17. If $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$, $x \neq 0$, then the roots of the equation is

- a. $\left\{ \pm 2, \frac{-3 \pm \sqrt{13}}{2} \right\}$ b. $\left\{ \pm 1, \frac{3 \pm \sqrt{13}}{2} \right\}$
 c. $\left\{ \pm 3, \frac{3 \pm \sqrt{11}}{2} \right\}$ d. None of these

Sol. b. $x^4 - 3x^3 - 2x^2 + 3x + 1 = 0$

On dividing throughout by x^2 and rearranging the terms, we get

$$\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 = 0 \quad \dots(i)$$

Put $x - \frac{1}{x} = y$

On squaring both sides, we get

$$\begin{aligned} x^2 + \frac{1}{x^2} - 2 = y^2 &\Rightarrow x^2 + \frac{1}{x^2} = y^2 + 2 \\ \Rightarrow (y^2 + 2) - 3y - 2 = 0 &\quad [\text{from Eq. (i)}] \\ \Rightarrow y^2 - 3y = 0 &\Rightarrow y(y - 3) = 0 \end{aligned}$$

$$\therefore y = 0 \text{ or } 3$$

For $y = 0$

$$\Rightarrow x - \frac{1}{x} = 0$$

$$\Rightarrow \frac{x^2 - 1}{x} = 0$$

$$\Rightarrow x^2 = 1$$

$$\therefore x = \pm 1$$

For $y = 3$

$$\Rightarrow x - \frac{1}{x} = 3$$

$$\Rightarrow \frac{x^2 - 1}{x} = 3$$

$$\Rightarrow x^2 - 3x - 1 = 0$$

$$\Rightarrow x = \frac{3 \pm \sqrt{9 + 4}}{2}$$

$$\therefore x = \frac{3 \pm \sqrt{13}}{2}$$

So, $x = \left\{ \pm 1, \frac{3 \pm \sqrt{13}}{2} \right\}$ is the required solution.

Note An equation of degree n has n roots.

Word Problems Involving Quadratic Equation

In this section, we will discuss some problems based on practical applications of quadratic equation.

In such type of problems,

- We formulate a quadratic equation in variable x with the help of the given conditions.
- We then solve the quadratic equation to find the answer to the given problem.

- Sometimes it may happen that, out of the roots of the quadratic equation only one satisfies the given condition in the problem. So, the root which does not satisfy the condition of the problem will be rejected.

EXAMPLE 18. A positive number, when increased by 10 equals 200 times its reciprocal. What is number?

- a. 100 b. 10 c. 20 d. 200

Sol. b. Let the positive number be x .

Then, according to the question,

$$\begin{aligned} x + 10 &= \frac{200}{x} \Rightarrow x^2 + 10x = 200 \\ \Rightarrow x^2 + 10x - 200 &= 0 \Rightarrow x^2 + 20x - 10x - 200 = 0 \\ \Rightarrow x(x + 20) - 10(x + 20) &= 0 \Rightarrow (x - 10)(x + 20) = 0 \\ \therefore x &= 10, -20 \end{aligned}$$

But $x \neq -20$, since x is a positive number.

So, the required number is 10.

EXAMPLE 19. Out of group of swans, $7/2$ times the square root of the number are swimming in the pool. While the two remaining are playing outside the pool. What is the total number of swans?

- a. 4 b. 8 c. 12 d. 16

Sol. d. Let total number of swans be x .

Number of swans swimming in the pool $= \frac{7}{2}\sqrt{x}$

Remaining swans $= 2$

$$\text{By given condition, } \frac{7}{2}\sqrt{x} + 2 = x \Rightarrow \frac{7}{2}\sqrt{x} = x - 2$$

On squaring both sides, we get $\frac{49}{4}x = x^2 + 4 - 4x$

$$\Rightarrow 4x^2 - 65x + 16 = 0, \quad 4x^2 - 64x - x + 16 = 0$$

$$\Rightarrow 4x(x - 16) - 1(x - 16) = 0 \Rightarrow (x - 16)(4x - 1) = 0$$

$$\therefore x = 16 \quad \left[\because x \neq \frac{1}{4} \right]$$

Thus, total number of swans $= 16$

INEQUALITIES

Two real numbers or two algebraic expressions related by the symbols $>$, $<$, $\leq$ or $\geq$ form inequality. Here, the symbols $<$ (less than), $>$ (greater than), $\leq$ (less than or equal) and $\geq$ (greater than or equal) are known as inequality signs.

e.g. $5 < 7$, $x \leq 2$, $x + y \geq 11$ are all inequalities.

Linear Inequalities

A linear inequality is an inequality which involves a linear function i.e. each variable occurs in first degree only and there is no term involving the product of the variables.

e.g. $ax + b \leq 0$, $ax + by + c > 0$, $ax \leq 4$.